

Day 5: IP Geolocation, ASN Mapping & Telephony OSINT Integration

Key Milestone: Threat Intelligence Enrichment & Telephony OSINT

Sprint Focus: Execution of core deliverables for Day 5 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: Geo-ASN Enrichment Service, DMARC Alignment Engine, Global Threat Map Component.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | Key Modules: NLP/AI Engine, Explainability, Content Risk scoring

Day 5 Execution: Implemented Bayesian Content Risk Scorer computing statistical word weights and adversarial token manipulation flags.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | Key Modules: Header Forensics, Mail Path, Infrastructure Risk

Day 5 Execution: Built DMARC policy evaluator inspecting alignment between RFC 5322 From and RFC 5321 Return-Path headers (``p=reject``, ``p=quarantine``).
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | Key Modules: IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 5 Execution: Integrated IP-API, MaxMind GeoLite, and BGP routing lookups to map originating ISP, Autonomous System Number (ASN), and country.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | Key Modules: Graph Engine, Campaign Detection, Similar Attack Detection

Day 5 Execution: Developed Campaign Detection Engine: clustering recurring email attacks into unified campaign IDs based on cryptographic fingerprinting.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | Key Modules: Risk Scoring, Case Management, Response Recommendations

Day 5 Execution: Added evidence packaging module generating cryptographic SHA-256 hashes for all raw email headers, body texts, and attachment metadata.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | Key Modules: Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 5 Execution: Built the Geolocation World Map view rendering suspect origin coordinates, ISP details, and routing hops.

System Integration & Quality Verification

Cross-Module Integration: Enriched email payloads with IP geo-coordinates and displayed real-time origin mapping on the Cockpit dashboard.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 5 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 5 MILESTONE SIGNED OFF